

TKE215203 Power System Dynamic and Stability

Chapter 7 : Linearized Models of Single-Machine to Infinite-Bus Systems

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I. Introduction

II. SMIB Linearization

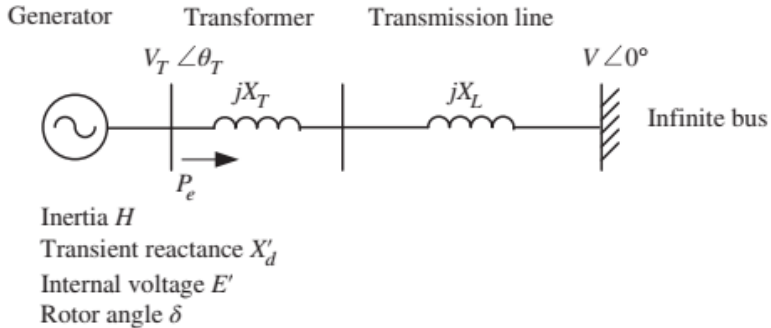
Section 1 :

Introduction

- Small-signal stability involves the system's ability to remain near an equilibrium point during small disturbances such as load or setpoint changes.
- Nonlinear simulation can indicate stability, but linear analysis offers more detailed insights including instability modes and damping margins.
- The key focus is on electromechanical oscillations caused by rotor interactions through network impedances.
- These oscillations typically range from 0.2 to 3 Hz and often have damping ratios below 10 percent.
- If damping is too low, small disturbances can cause persistent or overlapping oscillations.
- This chapter addresses how to model and analyze these dynamics using linear system models valid near operating points.
- Both analytical and numerical techniques are introduced to estimate frequency and damping of oscillatory modes.

Section 2 :

SMIB Linearization



The swing equations in state-variable form are:

$$\begin{aligned}\dot{\delta} &= \Omega\omega \\ 2H\dot{\omega} &= T_m - \frac{E'V \sin \delta}{X_{eq}} - D\omega\end{aligned}$$

where:

- δ : rotor angle
- ω : speed deviation (pu)
- $\Omega = 2\pi f$: conversion factor
- $X_{eq} = X'_d + X_T + X_L$

Steady-State Equilibrium

At steady-state, the equilibrium satisfies:

$$\omega = 0 \quad \text{and} \quad \frac{E' V \sin \delta_{ep}}{X_{eq}} = P_e = T_m$$

Thus, mechanical input equals electrical output and no speed deviation occurs.

Linearizing around $\delta = \delta_{ep}, \omega = 0$:

$$\Delta \dot{\delta} = \Omega \Delta \omega$$

$$2H\Delta \dot{\omega} = -K_s \Delta \delta - D\Delta \omega + \Delta T_m$$

where:

$$K_s = \frac{E' V \cos \delta_{ep}}{X_{eq}}$$

K_s is the synchronizing torque coefficient.

The linearized system can be written as:

$$\begin{bmatrix} \Delta \dot{\delta} \\ \Delta \dot{\omega} \end{bmatrix} = \begin{bmatrix} 0 & \Omega \\ -\frac{K_s}{2H} & -\frac{D}{2H} \end{bmatrix} \begin{bmatrix} \Delta \delta \\ \Delta \omega \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{2H} \end{bmatrix} \Delta T_m$$

The characteristic equation is:

$$\lambda^2 + \frac{D}{2H}\lambda + \frac{K_s\Omega}{2H} = 0$$

Solving the characteristic equation gives:

$$\lambda = \frac{-(D/2H) \pm \sqrt{(D/2H)^2 - 4(K_s\Omega/(2H))}}{2}$$

Assuming D is small:

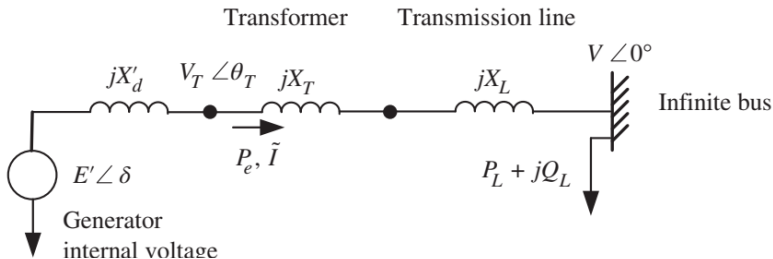
$$\lambda = \frac{1}{4H} \left(-D \pm j\sqrt{8HK_s\Omega - D^2} \right)$$

where:

$$\sigma = -\frac{D}{4H}, \quad \omega_s = \sqrt{\frac{K_s\Omega}{2H}}$$

σ is the damping in 1/sec, and ω_s is the oscillation frequency in rad/s.

Example: SMIB



The SMIB system operates at 50 Hz. Machine rated at 900 MVA with:

- Inertia $H = 3.5$ sec
- Damping $D = 1$ pu
- Transient reactance $X'_d = 0.24$ pu
- Transformer reactance $X_T = 0.15$ pu
- Line reactance $X_L = 0.06$ pu (system base 100 MVA)

Load flow:

$$\tilde{V}_T = 1.03 \angle 39.25^\circ \text{ pu}, P_e = 8.5 \text{ pu}, Q_e = 3.4338 \text{ pu}, \tilde{V} = 1.0 \angle 0^\circ \text{ pu}$$

Goal: Find the linearized model and its electromechanical mode.

Solution: Computation Steps

Generator current:

$$\tilde{I} = \frac{(P_e + jQ_e)^*}{\tilde{V}_T} = 8.5 + j2.6395 \text{ pu}$$

Machine internal voltage:

$$E' \angle \delta = \tilde{V}_T + jX'_d \tilde{I} = 1.1140 \angle 50.37^\circ \text{ pu}$$

Equivalent reactance:

$$X_{eq} = X_L + (X_T + X'_d) \times \frac{S_{base}}{S_{machine}} = 0.1033 \text{ pu}$$

Synchronizing torque coefficient:

$$K_s = \frac{E' V \cos \delta}{X_{eq}} = 5.8632$$

Linearized dynamic system:

$$\begin{bmatrix} \Delta \dot{\delta} \\ \Delta \dot{\omega} \end{bmatrix} = \begin{bmatrix} 0 & 314.16 \\ -0.1117 & -0.1429 \end{bmatrix} \begin{bmatrix} \Delta \delta \\ \Delta \omega \end{bmatrix} + \begin{bmatrix} 0 \\ 0.0159 \end{bmatrix} \Delta T_m$$

Characteristic polynomial:

$$\lambda^2 + 0.1429\lambda + 35.0948 = 0$$

Eigenvalues and Oscillatory Mode

The eigenvalues are:

$$\lambda_{1,2} = -0.0714 \pm j5.9237$$

Oscillatory mode frequency:

$$\omega_s = 5.9237 \text{ rad/sec} \quad f_{\text{osc}} = \frac{\omega_s}{2\pi} = 0.9425 \text{ Hz}$$

Damping:

$$\sigma = -0.0714 \text{ (1/sec)}$$

Observations on Local Mode Frequency

- ④ The frequency ω_s is inversely proportional to the square root of H and X_{eq} .
- ② As P_e (loading or power transfer) increases, ω_s decreases.
- ③ The synchronizing torque K_s is inversely proportional to X_{eq} and decreases as P_e (or δ) increases. Thus as K_s decreases, ω_s also decreases.